THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 3: **Metals and Non-metals** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** malleability	Malleability.
2	**(d)** gold	Gold is at the bottom of the series.
3	**(d)** haematite	Haematite is Fe_2O_3.
4	**(c)** ductility	Ductility.
5	**(d)** potassium	K is at the top.
6	**(a)** bauxite	Bauxite.
7	**(d)** lustre	Metallic lustre.
8	**(b)** iron	Iron rusts.
9	**(d)** copper glance	Cu_2S.
10	**(a)** sonorous	Sonorous.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Malleability: beaten into sheets (1); Ductility: drawn into wires (1).	1+1
12	Brass: Cu and Zn (1); Bronze: Cu and Sn (1).	1+1
13	Magnesium oxide (1); Equation $2\,Mg + O_2 \longrightarrow 2\,MgO$ (1).	1+1
14	Silver (1); Copper (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Lustre (1); Malleable and ductile (1); Good conductors of heat and electricity (1).	1+1+1
16	$2\,Na + 2\,H_2O \longrightarrow 2\,NaOH + H_2$ (2); Sodium hydroxide and hydrogen (1).	2+1
17	Copper: wires (1); Aluminium: foil (1); Iron: bridges and machines (1).	1+1+1
	OR: Dull appearance (1); Brittle and non-conductors (1); Low melting points (1).	1+1+1
18	$MgO + H_2O \longrightarrow Mg(OH)_2$ (2); Magnesium hydroxide is basic (1).	2+1
	OR: Mercury (1); Liquid at room temperature (1); Expands uniformly (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Lustre: metals lustrous, non-metals dull (1); Conduction: metals conduct (1); Malleability: metals malleable, non-metals brittle (1); Hardness: metals generally hard (1); Oxides: basic vs acidic (1).	1+1+1+1+1
	OR: Brass: Cu + Zn (1); Bronze: Cu + Sn (1); Solder: Pb + Sn (1); Stainless steel: Fe, Ni, Cr (1); Amalgam: Hg alloy (1).	1+1+1+1+1
20	$2\,Na + 2\,H_2O \longrightarrow 2\,NaOH + H_2$ (1); Na reacts violently (1); $Mg + 2\,H_2O \longrightarrow Mg(OH)_2 + H_2$ (hot water) (1); $3\,Fe + 4\,H_2O \longrightarrow Fe_3O_4 + 4\,H_2$ (steam) (1); Hydrogen is released (1).	1+1+1+1+1
	OR: Na (2,8,1) loses 1 electron (1); Cl (2,8,7) gains 1 electron (1); Na^+ and Cl^- attract (1); Ionic bond (1); $Na + Cl \longrightarrow NaCl$ (1).	1+1+1+1+1

Mistakes to avoid:

- Storing sodium or potassium in water; they are stored under kerosene because they react violently with water and air.
- Confusing **roasting** (sulphide ore, excess air) with **calcination** (carbonate ore, limited air).
- Writing the product of iron with steam as Fe_2O_3; it is Fe_3O_4 and hydrogen gas is released.
- Thinking that all metals are hard solids and all non-metals are non-conductors; Na is soft, Hg is a liquid and graphite conducts.
- Saying that an ionic solid conducts electricity; it conducts only when molten or in aqueous solution.
- Writing that copper can displace hydrogen from dilute acids or iron from $FeSO_4$; a less reactive metal cannot displace a more reactive one.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 3: **Metals and Non-metals** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** $Fe_2O_3 \cdot x\,H_2O$	Hydrated iron oxide.
2	**(c)** zinc blende	ZnS.
3	**(d)** sulphur	Sulphur is a non-metal.
4	**(b)** basic copper carbonate	Copper is attacked by moist CO_2.
5	**(c)** gangue	Gangue.
6	**(b)** magnesium	Mg is a metal.
7	**(a)** silver sulphide	Reaction with H_2S.
8	**(b)** an ore	Definition.
9	**(d)** silver	Silver then copper.
10	**(b)** an alloy	Alloy.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Mercury (1); Bromine (1).	1+1
12	Minerals from which metals are extracted (1); Bauxite and haematite (1).	1+1
13	Sulphur: matches, gunpowder (1); Carbon: fuel, pencil (1).	1+1
14	Corrosion of iron in moist air (1); $Fe_2O_3 \cdot x\,H_2O$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Metals are lustrous; non-metals dull (1); Metals conduct; non-metals do not (1); Metals malleable; non-metals brittle (1).	1+1+1
16	Brass: utensils (1); Bronze: statues (1); Solder: joining wires (1).	1+1+1
17	Light and good conductor of heat (1); Malleable (1); Does not corrode due to oxide layer (1).	1+1+1
	OR: Most reactive: Na (1); Na > Mg > Zn > Fe (1); Cu is the least reactive (1).	1+1+1
18	It reacts vigorously with air and water (1); Kerosene prevents contact (1); Potassium (1).	1+1+1
	OR: Metal oxides are basic, e.g. CaO (1); Non-metal oxides are acidic, e.g. CO_2 (1); Amphoteric: Al_2O_3 (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	$Mg + 2\,HCl \longrightarrow MgCl_2 + H_2$ (1); $Zn + 2\,HCl \longrightarrow ZnCl_2 + H_2$ (1); Cu: no reaction (1); Mg is the fastest (1); Order Mg > Zn > Cu (1).	1+1+1+1+1
	OR: Solid, hard and brittle (1); High melting points (1); Soluble in water (1); Conduct electricity in solution (1); Do not conduct in solid state (1).	1+1+1+1+1
20	$Fe + CuSO_4 \longrightarrow FeSO_4 + Cu$ (1); Green solution, brown deposit (1); $Zn + CuSO_4 \longrightarrow ZnSO_4 + Cu$ (1); Blue fades (1); Zn and Fe are more reactive than Cu (1).	1+1+1+1+1
	OR: Iron: structures (1); Copper: wires (1); Carbon: fuel (1); Sulphur: matches (1); Chlorine: disinfecting water (1).	1+1+1+1+1

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- Writing the product of iron with steam as Fe_2O_3; it is Fe_3O_4 and hydrogen gas is released.
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ANSWER KEY & MARKING SCHEME SET 3

Chapter 3: **Metals and Non-metals** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** roasting	Roasting.
2	**(d)** graphite	Graphite is an allotrope of carbon.
3	**(d)** amalgam	Mercury alloys are amalgams.
4	**(b)** calcination	Calcination.
5	**(d)** diamond	Diamond is the hardest.
6	**(d)** zinc	Brass: Cu + Zn.
7	**(c)** refining	Refining.
8	**(d)** sodium	It can be cut with a knife.
9	**(d)** zinc	Zinc coating.
10	**(a)** oxide	Metal oxides form.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Roasting: heating sulphide ore in excess air (1); Calcination: heating carbonate ore in limited air (1).	1+1
12	Homogeneous mixture of metals (1); Brass (1).	1+1
13	Sodium (1); Potassium (or calcium) (1).	1+1
14	Coating of zinc on iron (1); To prevent rusting (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$Zn + 2\,HCl \longrightarrow ZnCl_2 + H_2$ (2); Hydrogen (1).	2+1
16	Mineral: naturally occurring compound (1); Ore: mineral from which metal is extracted profitably (1); Gangue: impurity in ore (1).	1+1+1
17	Hydrogen (1); Burns with a pop sound (1); $Zn + H_2SO_4 \longrightarrow ZnSO_4 + H_2$ (1).	1+1+1
	OR: $Fe + CuSO_4 \longrightarrow FeSO_4 + Cu$ (2); Blue to pale green (1).	2+1
18	Painting (1); Oiling and greasing (1); Galvanisation (1).	1+1+1
	OR: Compounds formed by transfer of electrons from metal to non-metal (1); NaCl (1); $MgCl_2$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Corrosion: slow destruction of metals (1); Rusting of iron (1); Needs air and moisture (1); Prevent by painting (1); Galvanising and oiling (1).	1+1+1+1+1
	OR: Acidic: CO_2 (1); Basic: CaO (1); Neutral: CO (1); Amphoteric: Al_2O_3 (1); Amphoteric reacts with acid and base both (1).	1+1+1+1+1
20	Concentration of ore (1); Roasting or calcination (1); Reduction with carbon (1); $ZnO + C \longrightarrow Zn + CO$ (1); Refining (1).	1+1+1+1+1
	OR: Anode: impure copper (1); Cathode: pure copper strip (1); Electrolyte: acidified copper sulphate (1); Pure copper deposits on the cathode (1); Impurities fall as anode mud (1).	1+1+1+1+1

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